

SLEEVEDRIFT · DESIGN DECISIONS REGISTER

SleeveDrift design decisions register

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	SVD-DEC-001	Version	0.1
Status	Draft	Date	2026-10-03
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/sleevedrift		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-03	Amish Chadha	Register opened; design decisions made under Amish's pre-approvals of 2026-10-03; three requirement decisions proposed for Amish

SleeveDrift design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` or in `docs/REVIEW.md`; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list decisions.

Safety: Several decisions below set SleeveDrift's safety case for people working inside a rescue pipe (overload limit, gas monitoring, air, ring fixing, casualty evacuation). Each took the conservative option; the evidence that would relax it is in SVD-DDR-001, Table 1.

■ Open decisions

Three requirements are not met or are at risk on paper. They are put to Amish with options and a recommendation (full statement in `docs/REVIEW.md`, TRL 3, "Decisions for Amish"). Each is **Proposed, awaiting Amish**.

#	To decide	Options (effect on the requirement, cost, mass)	Recommendation	What it affects in the build	Source
1	R3 spoil haulage: 0.22 m ³ /h by hand	A. Keep hand cranking only: 0.22 m ³ /h, no cost, no mass. B. A right-angle drill with a winch bit on each winch at the portal: about 0.42 m ³ /h, about USD 700, about 7 kg. C. B plus six sledges, two people tipping and spoil pre-bagged at the face: about 0.71 m ³ /h, about USD 940, about 22 kg	B: it doubles the haul and moves it past the hand-dug face rate seen at Silkyara (about 0.27 m ³ /h), keeps hand cranking as the fallback and adds nothing inside the pipe	Portal station only: drill, winch bits, batteries	SVD-CAL-001, D; REVIEW TRL 3
2	R9 shift durability: rope abrasion unknown	A. Inspect both ropes every shift against a wear gauge and carry a spare set: no change to R9 on paper but a 20 min swap if one wears, USD 237, 15 kg. B. Rope with an abrasion-resistant coated cover: likely longer life, about USD 300 more, about 3 kg more. C. Low rollers every 10 m to lift the ropes off the floor: least wear, about USD 500, about 25 kg, and parts in the crawl lane	A, with the TRL 4 endurance trial deciding whether B is needed	Ropes and the shift routine; nothing to make	SVD-CAL-001, H; REVIEW TRL 3
3	R1 stretch: 600 mm pipe not designed	A. Remove the 600 mm stretch from this design: no cost, no mass. B. Design a 600 mm variant now (560 mm ring in three segments, 220 mm sledges, 150 mm duct): about USD 900 more for a second ring set, about 70 kg. C. Keep it as a stretch and decide when the co-design partner confirms whether 600 mm pipes are used for rescue: no cost now	C: the sizes are parameters in the model, and the partner's answer decides whether a second ring is worth making	Nothing in the first prototype	SVD-CAL-001, A; REVIEW TRL 3

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	Pipe bore, wall and internal weld beads of the pipe used for trials	The ring is 760 mm OD with 20 mm clearance; a bead or a smaller bore changes the fit	SVD-CAL-001, A; R1
2	Winch power ratios, base bolt pattern and line rating	Handle forces assume ratios of about 13 and 40; the trestle top plates are drilled to the base	SVD-CAL-001, C; BOM line 17
3	Release load of the breakaway swivel pins bought	Sets the 2.5 kN limit that every part is sized on	SVD-CAL-001, C; R11
4	Sheave bore, bush and rating	The pin is 25 mm; the sheave must be rated at least 1,000 kg	BOM line 6
5	Friction of the jacking pads on the actual pipe surface	The ring's hold against the sheave pull assumes 0.20	SVD-CAL-001, C8
6	Rope breaking strength and splice efficiency	The factor of 6.5 assumes 18 kN and 90 % at the splice	SVD-CAL-001, C6
7	Blower curve at 525 Pa	The air figures assume 0.27 m ³ /s at the blower	SVD-CAL-001, F; R5
8	Holding force of the magnet hangers on painted or rusty pipe	They carry the duct and the cable	BOM line 22

■ Value engineering

Value-engineering target: USD 5,000 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 9,867.50 (USD 4,867.50 over the target). Main cost drivers and savings worth trying:

- The two self-tailing winches (USD 2,300) are the largest line; reconditioned winches of the same size are often about half the new price.
- The four-gas monitors (USD 1,100), the blower and duct (USD 1,515) and the telephones (USD 860) are kit that rescue services usually already hold; borrowing them for trials removes about USD 3,500 from the prototype cost without changing the design.
- The made steel (ring, trestles, sledges) is under USD 1,500; plate rolling is the main cost there.
- Savings not worth taking: the breakaway swivels, rope and slings carry the safety case and stay at rated grades.

■ Decisions made

Pre-approval quotes: Q1, Amish 2026-10-03: "start with the first 14 repos from the list of 29 projects. I pre-approve the batch runs along with any recommendations you come up with. I also accept any cost overruns or variations from the assumed scope cost." Q2, Amish 2026-10-03: "Proceed with the remaining 15 scaffolds" (under the same pre-approval).

Date	Decision	Decided by	Record
2026-10-03	Design-case pipe 800 x 10 mm, 780 mm bore	Amish, Q1 and Q2	SVD-DDR-001, item 1
2026-10-03	Ring held by jacking screws with swivel pads; nothing drilled or welded to the pipe	Amish, Q1 and Q2	SVD-DDR-001, item 2
2026-10-03	Shuttle train between a pull rope and a tail rope, in place of a circulating loop	Amish, Q1 and Q2	SVD-DDR-001, item 3
2026-10-03	Two two-speed self-tailing winches at the portal; SiltHaul capstan not used	Amish, Q1 and Q2	SVD-DDR-001, items 4, 12
2026-10-03	Breakaway swivels at 2.5 kN at both ends of the train; requirement R11 added	Amish, Q1 and Q2	SVD-DDR-001, item 5
2026-10-03	Two four-gas monitors in the kit; requirement R12 added	Amish, Q1 and Q2	SVD-DDR-001, item 6
2026-10-03	Forcing ventilation through 200 mm layflat, about 15 m³/min at the face	Amish, Q1 and Q2	SVD-DDR-001, item 7
2026-10-03	Sound-powered telephones with a pull-cord bell	Amish, Q1 and Q2	SVD-DDR-001, item 8
2026-10-03	Bought roll-up stretcher with a made spreader bar; casualty hauled hand over hand through winch A	Amish, Q1 and Q2	SVD-DDR-001, items 9, 10
2026-10-03	First co-design candidate to approach: India's National Disaster Response Force with the Jhansi hand miners (not agreed)	Amish, Q1 and Q2	SVD-DDR-001, item 11
2026-10-03	CalRig as the first candidate rig for the hood test and proof loads; HatchSide as the first reference for air and gas rules	Amish, Q1 and Q2	SVD-DDR-001, item 12
2026-10-03	budget_usd kept at 5,000 as a value-engineering target	Amish, Q1 ("I also accept any cost overruns or variations from the assumed scope cost.")	SVD-DDR-001, item 13
2026-10-03	Design for construction: the twelve changes of SVD-DDR-002	Amish, Q1 and Q2	SVD-DDR-002
2026-10-03	Appearance model additions for renders: pipe window and platform cropped, a 1.75 m mannequin beside the pipe	Amish, Q1 and Q2	docs/ REVIEW.md, TRL 3